

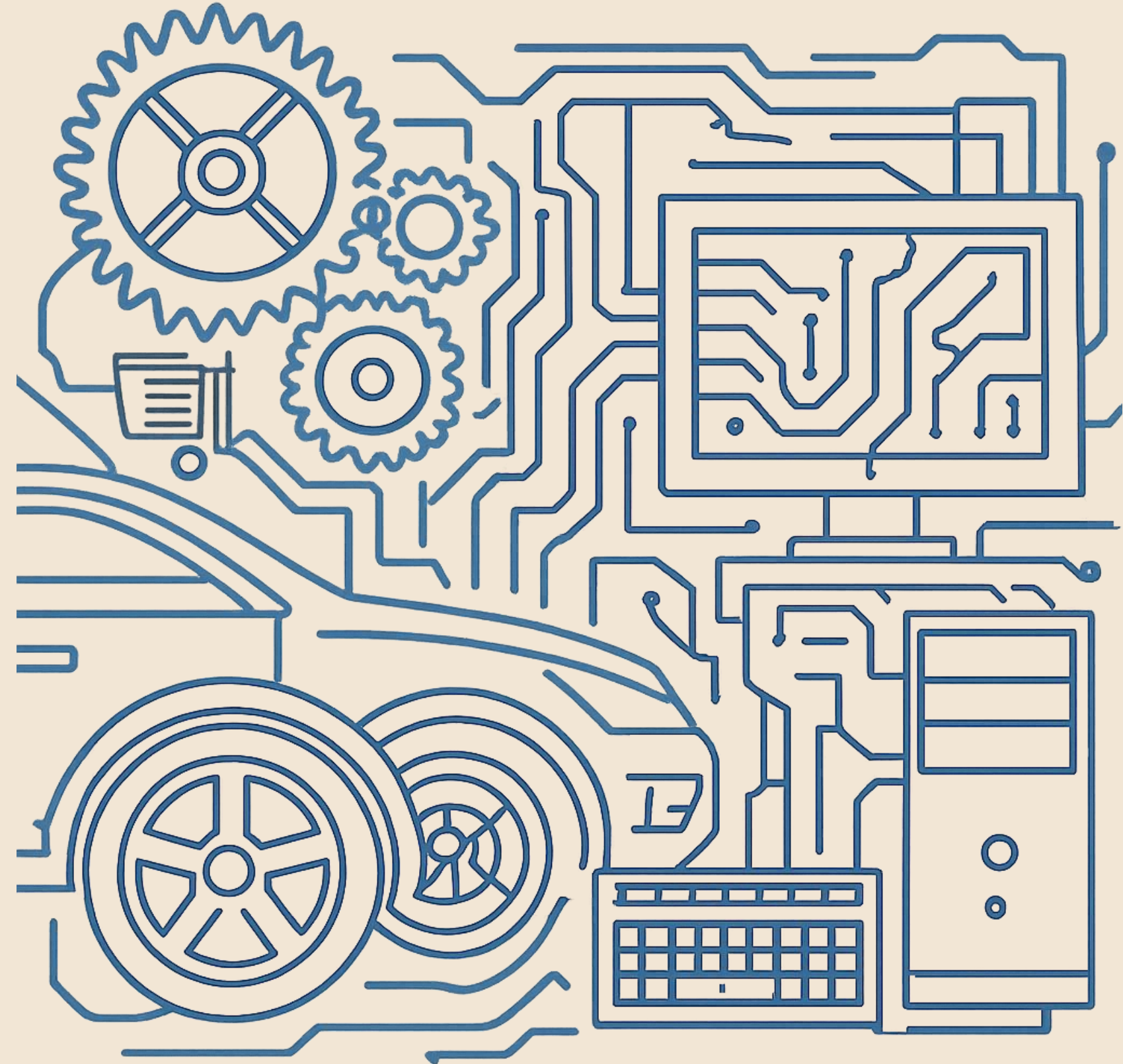


American
University of Beirut

VIP 201A - Presentation

Prediction of Driver Aggressiveness
Level via Driving Behavior

Hadi Al Shmaissani





American
University of Beirut

Agenda

01

**Problem
Identification**

02

Motivation

03

**Write your
agenda
point**

04

**Write your
agenda
point**



Process

- The Flaw: Autonomous vehicles struggle to anticipate aggressive human drivers.
- The Action: Analyze dynamic behaviors like speed, acceleration, and lane waviness.
- The Goal: Predict and quantify driver intent by computing an Aggressiveness Index (AI)

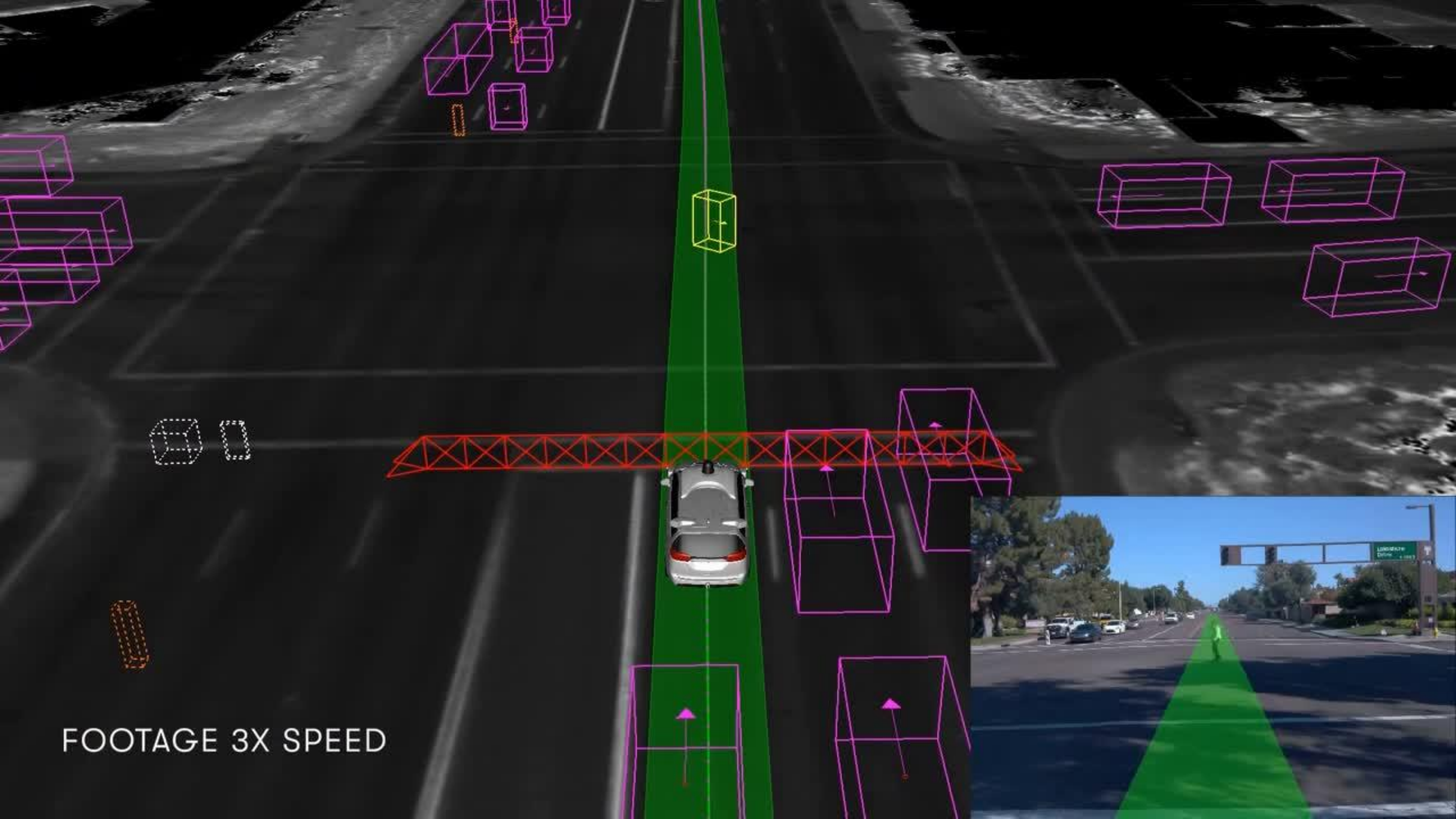


System

- The Gap: Current systems lack real-time classification for complex driving behaviors.
- The Tech: Combine computer vision for cue extraction with regression models for AI calculation.
- The Test: Validate the models using scripted aggressive behaviors in CARLA or SUMO.

Problem Identification







Research Background



Industry Issues

Autonomous systems struggle to anticipate unpredictable human actors. The core challenge is developing a way to predict and quantify the aggressiveness of surrounding drivers



Efficiency

The solution requires rapid analysis of dynamic vehicle behaviors, including speed, acceleration, proximity, and waviness. The system must be optimized for low real-time prediction latency.



Productivity

Key deliverables include building a categorized dataset of driving behaviors. This will power a robust mathematical or machine-learning model to compute an Aggressiveness Index (AI).





Objectives

Lorem ipsum dolor sit amet,
consectetur adipiscing elit.
Suspendisse viverra augue quis nulla
posuere varius.



01 Process

Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Suspendisse viverra augue quis
nulla posuere varius. Ut maximus magna vel
turpis hendrerit, quis sagittis felis viverra. Nunc
ligula felis, dapibus sed vulputate in, ullamcorper
vitae ligula. Fusce eleifend sagittis mi, pulvinar
blandit mauris cursus vitae.



02 Supporting

Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Suspendisse viverra augue quis
nulla posuere varius. Ut maximus magna vel
turpis hendrerit, quis sagittis felis viverra. Nunc
ligula felis, dapibus sed vulputate in, ullamcorper
vitae ligula. Fusce eleifend sagittis mi, pulvinar
blandit mauris cursus vitae.



Scope of Study

01

System

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Suspendisse viverra augue quis nulla posuere varius. Ut maximus magna vel turpis hendrerit, quis sagittis felis viverra. Nunc ligula felis, dapibus sed vulputate in, ullamcorper vitae ligula. Fusce eleifend sagittis mi, pulvinar blandit mauris cursus vitae.

02

Analysis

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Suspendisse viverra augue quis nulla posuere varius. Ut maximus magna vel turpis hendrerit, quis sagittis felis viverra. Nunc ligula felis, dapibus sed vulputate in, ullamcorper vitae ligula. Fusce eleifend sagittis mi, pulvinar blandit mauris cursus vitae.





Research Model



Input

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Suspendisse viverra augue quis nulla posuere varius. Ut maximus magna vel turpis hendrerit, quis sagittis felis viverra. Nunc ligula felis, dapibus sed vulputate in, ullamcorper vitae ligula.

Process

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Suspendisse viverra augue quis nulla posuere varius. Ut maximus magna vel turpis hendrerit, quis sagittis felis viverra. Nunc ligula felis, dapibus sed vulputate in, ullamcorper vitae ligula.

Output

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Suspendisse viverra augue quis nulla posuere varius. Ut maximus magna vel turpis hendrerit, quis sagittis felis viverra. Nunc ligula felis, dapibus sed vulputate in, ullamcorper vitae ligula.

Control

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Suspendisse viverra augue quis nulla posuere varius. Ut maximus magna vel turpis hendrerit, quis sagittis felis viverra. Nunc ligula felis, dapibus sed vulputate in, ullamcorper vitae ligula.



Data Collection

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Suspendisse viverra augue quis nulla posuere varius. Ut maximus magna vel turpis hendrerit, quis sagittis felis viverra. Nunc ligula felis, dapibus sed vulputate in, ullamcorper vitae ligula. Fusce eleifend sagittis mi, pulvinar blandit mauris cursus vitae.

Analysis

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Suspendisse viverra augue quis nulla posuere varius. Ut maximus magna vel turpis hendrerit, quis sagittis felis viverra. Nunc ligula felis, dapibus sed vulputate in, ullamcorper vitae ligula. Fusce eleifend sagittis mi, pulvinar blandit mauris cursus vitae.

Methodology

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Suspendisse viverra augue quis nulla posuere varius.





Results

Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Suspendisse viverra augue quis nulla posuere varius.

× × × ×
× × × ×
× × × ×



Efficiency

Lorem ipsum dolor sit amet,
consectetur adipiscing elit.
Suspendisse viverra augue quis nulla
posuere varius. Ut maximus magna vel
turpis hendrerit, quis sagittis felis
viverra. Nunc ligula felis, dapibus sed
vulputate in, ullamcorper vitae ligula.



Improvement

Lorem ipsum dolor sit amet,
consectetur adipiscing elit.
Suspendisse viverra augue quis nulla
posuere varius. Ut maximus magna vel
turpis hendrerit, quis sagittis felis
viverra. Nunc ligula felis, dapibus sed
vulputate in, ullamcorper vitae ligula.



Findings

Lorem ipsum dolor sit amet,
consectetur adipiscing elit.
Suspendisse viverra augue quis nulla
posuere varius. Ut maximus magna vel
turpis hendrerit, quis sagittis felis
viverra. Nunc ligula felis, dapibus sed
vulputate in, ullamcorper vitae ligula.

× × × × ×
× × × × ×



Discussion

Lorem ipsum dolor sit amet,
consectetur adipiscing elit.
Suspendisse viverra augue quis nulla
posuere varius.



Interpretation

Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Suspendisse viverra augue quis nulla posuere varius. Ut
maximus magna vel turpis hendrerit, quis sagittis felis
viverra. Nunc ligula felis, dapibus sed vulputate in,
ullamcorper vitae ligula. Fusce eleifend sagittis mi, pulvinar
blandit mauris cursus vitae.



Impact

Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Suspendisse viverra augue quis nulla posuere varius. Ut
maximus magna vel turpis hendrerit, quis sagittis felis
viverra. Nunc ligula felis, dapibus sed vulputate in,
ullamcorper vitae ligula. Fusce eleifend sagittis mi, pulvinar
blandit mauris cursus vitae.



Conclusion



lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam quis libero pharetra, venenatis erat eu, fermentum ante. Nullam in dictum odio. Morbi luctus efficitur lorem vitae commodo. Orci varius natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Nulla feugiat risus ante, id ultricies est accumsan facilisis. Praesent molestie leo porttitor ante tempor, sed convallis tortor euismod. Suspendisse at nibh massa. Aenean at nisi eget augue commodo placerat.

Ut ex nulla, volutpat vitae est vel, ornare pharetra diam. Cras sit amet euismod magna. Praesent lacinia mi a enim luctus malesuada. Sed dignissim vehicula ligula, sit amet cursus mi vestibulum quis.





Thank You

Questions & Discussion

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Suspendisse viverra augue quis nulla posuere varius. Ut maximus magna vel turpis hendrerit, quis sagittis felis viverra. Nunc ligula felis, dapibus sed vulputate in, ullamcorper vitae ligula. Fusce eleifend sagittis mi, pulvinar blandit mauris cursus vitae.

